# T3 — Organisms Exchange Substances

## 3.1 Surface Area to Volume Ratio

**Surface area to volume ratio (SA:V):** As organisms increase in size, their volume increases more rapidly than their surface area (volume increases as the cube of linear dimensions, surface area as the square). This means larger organisms have a progressively smaller SA:V ratio, making simple diffusion across the body surface insufficient to supply all cells with nutrients and oxygen, or to remove waste products quickly enough.

Consequences of a small SA:V ratio in large, multicellular organisms:

- Diffusion distances become too great — the interior cells are too far from the external surface for diffusion alone to meet their metabolic demands.
- **Specialised exchange surfaces** are required — organs with adaptations to maximise the rate of exchange (e.g. lungs, gills, villi in the intestine, root hair cells).
- **Mass transport systems** are required — circulatory systems (blood) and ventilation systems (breathing) move substances rapidly between exchange surfaces and the body's cells, maintaining concentration gradients.

Features common to all effective exchange surfaces (remembered by the mnemonic **SLIM**):

- **S**urface area — large, to increase the rate of diffusion (e.g. alveoli provide ~70 $m^2$, villi/microvilli massively increase intestinal surface area)
- **L**ength of diffusion pathway — thin barrier, often just one or two cells thick (e.g. alveolar epithelium + capillary endothelium = ~0.5 micrometres total)
- **I**nternal medium maintained — good blood supply or ventilation maintains a steep concentration gradient across the exchange surface
- **M**oist surface — required for gas exchange so gases can dissolve before diffusing across cell membranes

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$$\text{Rate of diffusion} \propto (\text{Surface area} \times \text{Concentration difference}) / \text{Thickness of barrier}$$

(Fick's Law)

**Exam Tip:** Fick's Law is not just a formula to remember — use it to explain *why* each adaptation increases exchange rate. For example: "Alveoli have thin walls (reducing the thickness of the barrier) and a rich blood supply (maintaining a steep concentration gradient), both of which increase the rate of diffusion according to Fick's Law." Linking adaptations to the specific variables in Fick's Law is the key to full marks.

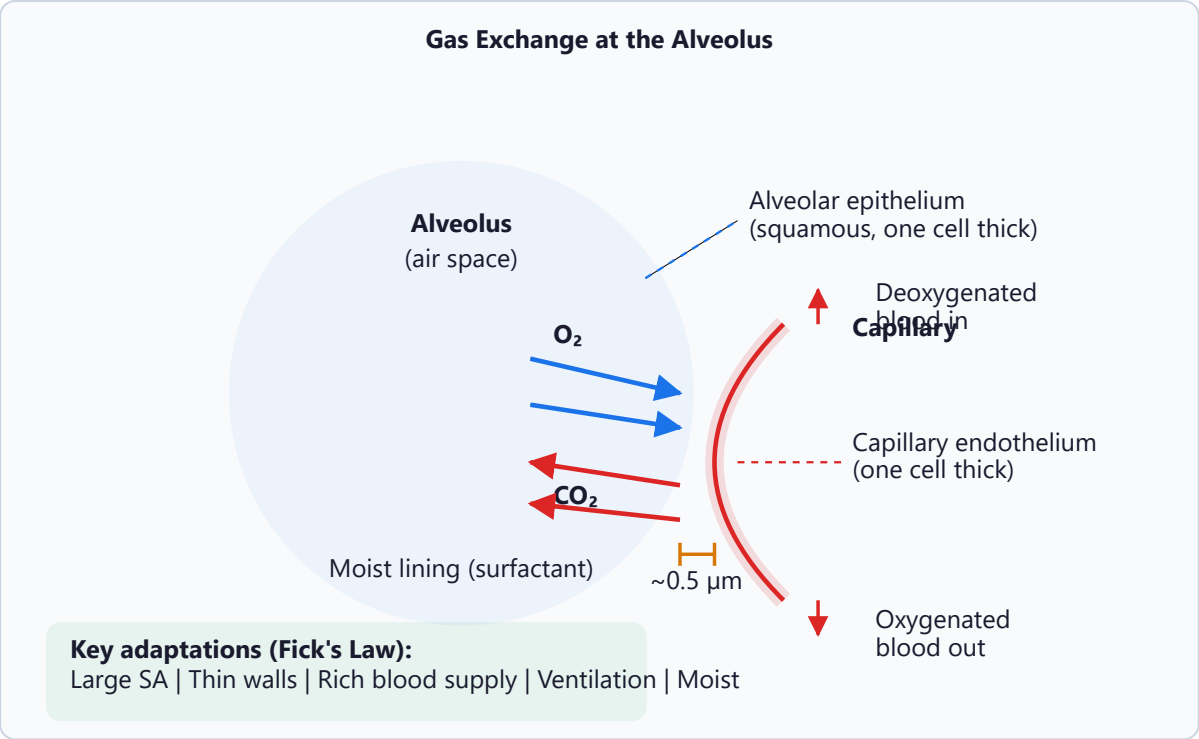
# 3.2 Gas Exchange

## Gas Exchange in Humans (Lungs)

The lungs provide a vast surface area for gas exchange between inhaled air and blood. The gas exchange surface is the **alveolar epithelium** — approximately 300 million alveoli provide a combined surface area of about 70 $m^2$ (roughly the area of half a tennis court).

Adaptations of alveoli for efficient gas exchange:

- **Enormous total surface area:** ~70 $m^2$ from ~300 million alveoli — maximises the area available for diffusion.
- **Extremely thin walls:** Alveolar epithelium is a single layer of squamous (flat) cells; the capillary endothelium is also one cell thick. Total diffusion distance is only ~0.5 micrometres, minimising the barrier thickness.
- **Rich blood supply:** Each alveolus is surrounded by a dense network of capillaries. Deoxygenated blood arriving has low $pO_2$ and high $pCO_2$; the alveolar air has high $pO_2$ and low $pCO_2$. The constant blood flow removes $O_2$ and delivers $CO_2$, maintaining steep concentration gradients.
- **Ventilation:** Breathing continuously refreshes the air in alveoli, maintaining the gradient. Without ventilation, the air in the alveoli would equilibrate with the blood and gas exchange would stop.
- **Moist lining:** Alveoli are lined with a thin layer of fluid (surfactant) — gases dissolve in this fluid before diffusing across the epithelium. Surfactant also reduces surface tension, preventing alveolar collapse.




## Ventilation Mechanism

Ventilation (breathing) is a mechanical process that moves air into and out of the lungs, maintaining concentration gradients for gas exchange. It relies on pressure changes in the thoracic cavity:

| Feature | Inspiration (Breathing in) | Expiration (Breathing out) |
|---|---|---|
| Diaphragm | Contracts and **flattens** (moves downward) | Relaxes and **domes upward** |
| External intercostal muscles | Contract — ribs pivot upward and outward | Relax — ribs fall downward and inward under gravity |
| Internal intercostal muscles | Relaxed | Contract during **forced** expiration only (not quiet breathing) |
| Thoracic volume | Increases | Decreases |
| Intrapulmonary pressure | Decreases (below atmospheric) — Boyle's Law | Increases (above atmospheric) |

| | | |
|---|---|---|
| Air movement | Air drawn **into** lungs (pressure gradient) | Air pushed **out** of lungs |
| Process type | **Active** (requires muscle contraction and ATP) | **Passive** during quiet breathing (elastic recoil of lungs); **active** during forced expiration |

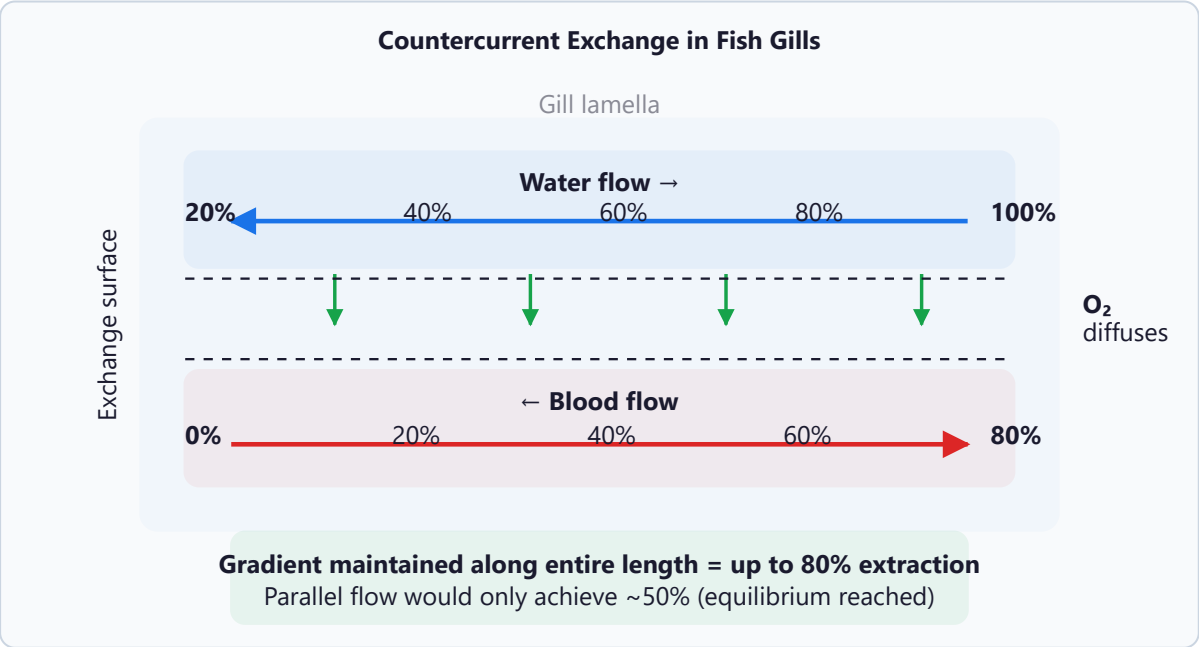
## Gas Exchange in Fish (Gills)

Fish extract dissolved oxygen from water, which contains far less $O_2$ than air (~1% vs ~21%). To compensate, fish gills use a highly efficient **countercurrent exchange system**.

Structure: Water enters through the mouth, passes over the gill filaments (each bearing many thin plate-like **lamellae** with a rich blood supply), and exits through the operculum (gill cover). Blood flows through capillaries in the lamellae in the **opposite direction** to water flow.

Why countercurrent is so efficient:

- Because blood and water flow in opposite directions, a concentration gradient for $O_2$ is maintained **along the entire length** of the lamella. At every point, the water always has a higher $pO_2$ than the adjacent blood, so $O_2$ continuously diffuses from water into blood.
- This allows up to **80% of the dissolved oxygen** to be extracted from the water — far more efficient than a parallel (concurrent) system, which would reach equilibrium at ~50% and then stop.



**Exam Tip:** The countercurrent system in fish gills is a very common exam question. Always explain *why* it is more efficient than parallel flow: because a concentration gradient is maintained along the entire length of the lamella, whereas in parallel flow, equilibrium would be reached partway along, and diffusion would stop. Draw a diagram showing the percentage $O_2$ values at each point if asked.

## Gas Exchange in Insects (Tracheal System)

Insects have an internal system of **air-filled tubes (tracheae)** that branch into progressively smaller tubes called **tracheoles**, delivering oxygen directly to respiring cells. This is fundamentally different from vertebrate systems — there is no oxygen-carrying blood pigment, and gas exchange occurs directly between the tracheole fluid/air and the cells.

- Air enters through **spiracles** (small pores on the body surface, with valves to reduce water loss).
- Tracheae are reinforced with rings of chitin to prevent collapse but remain flexible.
- Tracheoles penetrate between and into individual cells, providing an extremely short diffusion distance.
- The ends of tracheoles contain a small amount of fluid; during high metabolic activity, this fluid is drawn into cells by osmosis (due to accumulation of lactate), bringing air closer to the tissues.
- Ventilation is assisted by rhythmic body movements (abdominal pumping) that compress and expand the tracheae and air sacs, forcing air in and out.

This system is efficient for small organisms but limits body size — the reliance on diffusion over most of the pathway means insects cannot grow as large as vertebrates with circulatory systems. (Exceptions occurred in the Carboniferous period when atmospheric $O_2$ was ~35%, allowing giant dragonflies with 70 cm wingspans.)

## Gas Exchange in Plants (Leaves)

Gas exchange in plants occurs primarily through **stomata** — small pores mostly on the lower epidermis of leaves, each flanked by two kidney-shaped **guard cells** that control stomatal opening and closing.

- During the day: stomata are open for $CO_2$ uptake (photosynthesis) and $O_2$ release. Water vapour is also lost (transpiration).
- At night: stomata typically close (no photosynthesis), reducing water loss. $O_2$ is consumed and $CO_2$ is released by respiration only.

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- The spongy mesophyll has a large internal surface area with extensive air spaces between loosely packed cells, facilitating $CO_2$ diffusion to photosynthesising cells.
- Guard cells open stomata when $K^+$ ions are actively pumped in, lowering the water potential. Water enters by osmosis, the guard cells swell and become turgid, and their thickened inner walls cause them to bow apart, opening the pore.

# 3.3 Digestion and Absorption

## Digestion

**Digestion:** The breakdown of large, insoluble food molecules into smaller, soluble molecules that can be absorbed across the gut wall into the blood. There are two types: **mechanical digestion** (physical breakdown by teeth and churning, increasing surface area for enzyme action) and **chemical digestion** (hydrolysis of chemical bonds by digestive enzymes).

| Enzyme | Substrate | Products | Site of Action | Optimal pH |
|---|---|---|---|---|
| **Salivary amylase** | Starch (α-1,4 bonds) | Maltose | Mouth (and briefly oesophagus) | ~pH 7 (neutral) |
| **Pancreatic amylase** | Starch | Maltose | Duodenum (lumen) | ~pH 7-8 |
| **Maltase** | Maltose | Glucose + Glucose | Small intestine (brush border of epithelial cells) | ~pH 7 |
| **Sucrase** | Sucrose | Glucose + Fructose | Small intestine (brush border) | ~pH 7 |
| **Lactase** | Lactose | Glucose + Galactose | Small intestine (brush border) | ~pH 7 |

| | | | | |
|---|---|---|---|---|
| **Pepsin** (endopeptidase) | Proteins (internal peptide bonds) | Shorter polypeptides | Stomach | ~pH 2 (acidic) |
| **Trypsin** (endopeptidase) | Proteins (internal bonds) | Shorter polypeptides | Duodenum | ~pH 8 (alkaline) |
| **Exopeptidases** (e.g. dipeptidases, carboxypeptidase) | Terminal amino acids of polypeptides | Individual amino acids | Small intestine (brush border) | ~pH 7 |
| **Pancreatic lipase** | Triglycerides (ester bonds) | Fatty acids + Glycerol (+ monoglycerides) | Duodenum | ~pH 7-8 |

**Endopeptidases vs Exopeptidases:** Endopeptidases (pepsin, trypsin) cleave *internal* peptide bonds within a polypeptide chain, producing several shorter fragments and greatly increasing the number of free ends. Exopeptidases (dipeptidases, aminopeptidases, carboxypeptidases) then remove amino acids one at a time from these exposed ends. This two-stage strategy is efficient because endopeptidases create many fragments with many ends, providing multiple sites for exopeptidases to work simultaneously.

**Bile** is produced by the liver, stored and concentrated in the gall bladder, and released into the duodenum via the bile duct. Bile is not an enzyme — it has two key functions:

- **Emulsification:** Bile salts (amphipathic molecules) break large fat globules into many smaller droplets (an emulsion), massively increasing the surface area available for lipase to act on. Without emulsification, lipase can only access the surface of a large droplet, making fat digestion extremely slow.
- **Neutralisation:** Bile is alkaline (contains $NaHCO_3$), which neutralises the acidic chyme arriving from the stomach. This raises the pH to ~7-8, the optimum for pancreatic enzymes (amylase, trypsin, lipase).

## Absorption in the Ileum

The ileum (final section of the small intestine) is the primary site of nutrient absorption. It is highly adapted with **villi** (finger-like projections ~0.5-1 mm long) and **microvilli** (microscopic membrane folds on the apical surface of epithelial cells, forming the "brush border") that together increase the absorptive surface area by a factor of ~600.

Each villus has:

- **Single layer of columnar epithelial cells:** Thin barrier — short diffusion distance.
- **Microvilli (brush border):** Increase surface area by ~20-fold. Also contain membrane-bound digestive enzymes (maltase, sucrase, lactase, dipeptidases) for final-stage digestion.
- **Dense capillary network:** Maintains a steep concentration gradient for glucose and amino acids by rapidly carrying absorbed nutrients away in the blood (to the liver via the hepatic portal vein).
- **Central lacteal:** A blind-ended lymph vessel that absorbs lipids. Fatty acids and glycerol are recombined into triglycerides in the epithelial cells, packaged with cholesterol and proteins into **chylomicrons** (lipoprotein particles), and exported into the lacteal by exocytosis. Lipids travel via the lymphatic system rather than the blood because chylomicrons are too large for blood capillaries.
- **Numerous mitochondria:** Epithelial cells are packed with mitochondria to provide the ATP needed for active transport ($Na^+/K^+$ ATPase) and co-transport of glucose and amino acids.

**Glucose absorption** occurs by sodium-glucose co-transport (SGLT1) on the apical membrane and facilitated diffusion (GLUT2) on the basal membrane, as described in T2 section 2.3.

**Amino acid absorption** occurs by a similar co-transport mechanism — different amino acid transporters couple amino acid uptake to the sodium gradient.

## 3.4 Mass Transport — The Circulatory System

### Haemoglobin and Oxygen Transport

**Haemoglobin (Hb):** A globular protein with quaternary structure, found inside red blood cells. It consists of four polypeptide subunits (2 alpha, 2 beta chains), each containing an iron-containing **haem** prosthetic group that can reversibly bind one $O_2$ molecule. Fully saturated Hb carries 4 $O_2$ molecules (as oxyhaemoglobin, $HbO_8$).

$$\text{Hb} + 4\text{O}_2 \rightleftharpoons \text{HbO}_8 \text{ (oxyhaemoglobin) — loading in the lungs, unloading in tissues}$$

**Cooperative binding (positive cooperativity):** When the first $O_2$ molecule binds to one of the four haem groups, it causes a conformational change in the haemoglobin molecule that makes it easier for the second $O_2$ to bind, and easier still for the third and fourth. Conversely, when $O_2$ starts to dissociate, the

remaining $O_2$ molecules are released more readily. This produces the characteristic **S-shaped (sigmoid) oxygen dissociation curve**, rather than a simple hyperbolic curve.

The sigmoid curve means that:

- At the lungs (high $pO_2$, ~13 kPa): Hb is nearly 100% saturated — efficient loading.
- At resting tissues (moderate $pO_2$, ~5 kPa): Hb releases some $O_2$ (~25% of its load) — adequate for resting metabolic needs.
- At actively respiring tissues (low $pO_2$, ~2 kPa): Hb releases much more $O_2$ — the steep part of the curve means even a small drop in $pO_2$ causes a large release of $O_2$.

**The Bohr Effect:** In actively respiring tissues, increased $CO_2$ production leads to more carbonic acid formation ($CO_2 + H_2O \rightarrow H_2CO_3$, catalysed by carbonic anhydrase in red blood cells), which dissociates to release $H^+$ ions, lowering local pH. The $H^+$ ions bind to haemoglobin, causing a conformational change that **reduces its affinity for oxygen**. This means haemoglobin releases $O_2$ more readily to the tissues that need it most. On the dissociation curve, this is seen as a **shift to the right**.

## The Heart

The heart is a muscular double pump with four chambers. The **right side** pumps deoxygenated blood to the lungs via the pulmonary artery (pulmonary circulation). The **left side** pumps oxygenated blood to the body via the aorta (systemic circulation). The two sides are completely separated by the **septum**, preventing mixing of oxygenated and deoxygenated blood.

| Feature | Significance |
|---|---|
| Left ventricle wall much thicker than right | Generates much higher pressure needed to pump blood around the entire body (systemic circulation). The right ventricle only needs to pump blood the short distance to the lungs. |
| Atria have thin walls | Only need to push blood a short distance downward into the ventricles below — do not need to generate high pressure. |
| Atrioventricular (AV) valves: tricuspid (right) and bicuspid/mitral (left) | Prevent backflow of blood from ventricles into atria during ventricular systole. Close when ventricular pressure exceeds atrial pressure. |

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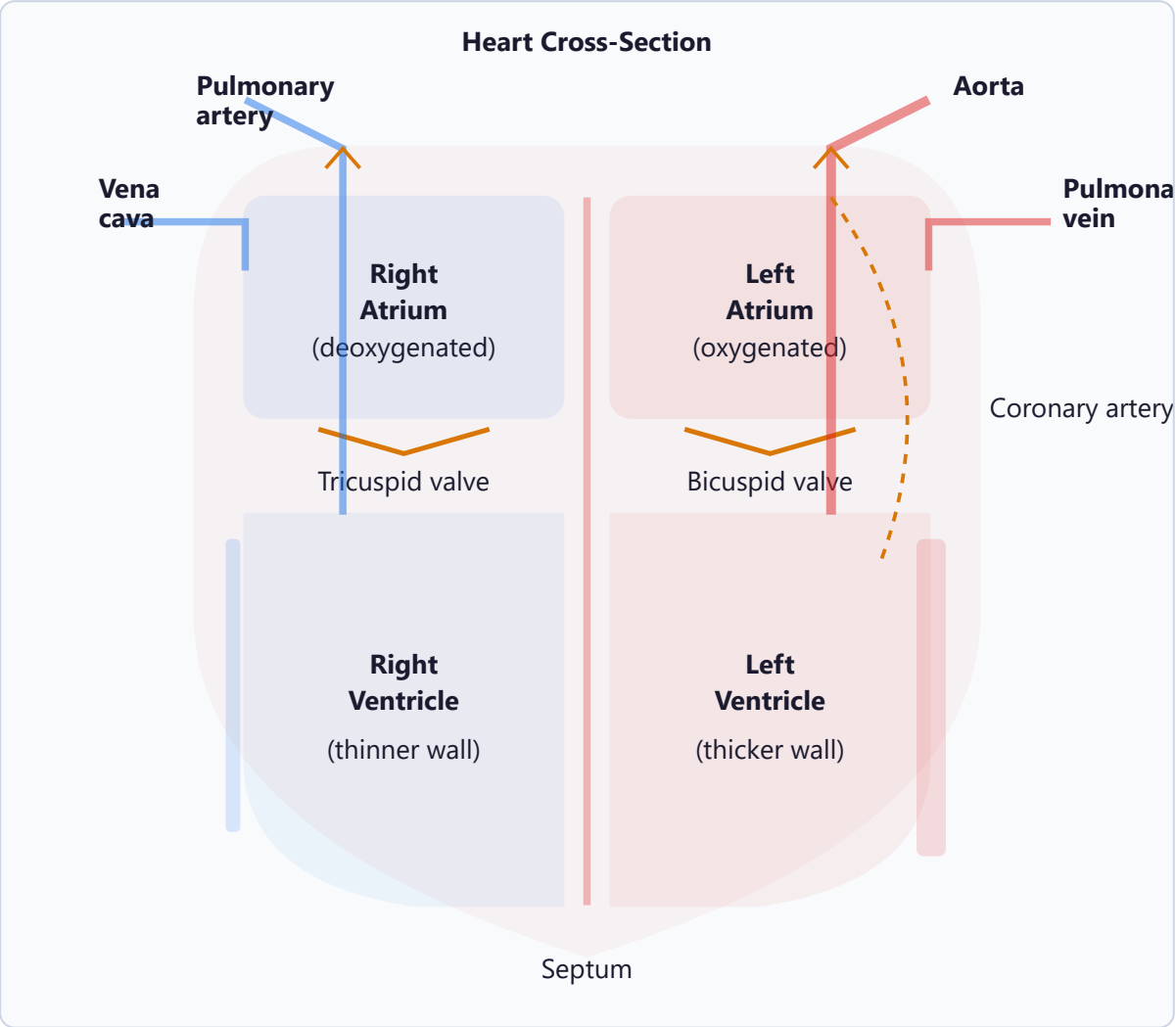
| | |
|---|---|
| Semilunar valves: aortic and pulmonary | Prevent backflow of blood from the aorta/pulmonary artery back into the ventricles during diastole. Close when arterial pressure exceeds ventricular pressure. |
| Coronary arteries (branch from base of aorta) | Supply the heart muscle (myocardium) itself with oxygenated blood and nutrients. Blockage causes myocardial infarction (heart attack). |




## Cardiac Cycle

The cardiac cycle is the sequence of events in one complete heartbeat (~0.8 seconds at rest, ~75 bpm):

1. **Atrial systole (~0.1 s):** Both atria contract simultaneously, raising atrial pressure above ventricular pressure. Blood is pushed through the open AV valves into the ventricles (topping them up — ventricles are already ~80% full from passive filling during diastole).

2. **Ventricular systole (~0.3 s):** Both ventricles contract simultaneously. Ventricular pressure rapidly exceeds atrial pressure, causing AV valves to close (producing the first heart sound — "lub"). Pressure continues to rise until it exceeds arterial pressure, forcing semilunar valves open. Blood is ejected into the aorta (left) and pulmonary artery (right).
3. **Diastole (~0.4 s):** All chambers relax. Ventricular pressure drops below arterial pressure — semilunar valves close (producing the second heart sound — "dub"), preventing backflow. Ventricular pressure drops below atrial pressure — AV valves open and blood flows passively from veins into atria and through into ventricles.

**The heart is myogenic** — the cardiac cycle is initiated by the heart's own pacemaker, not by nervous stimulation:

1. The **SAN (sinoatrial node)** in the wall of the right atrium generates spontaneous electrical impulses at a regular rate. These impulses spread as a wave of depolarisation across both atrial walls, causing **atrial systole**.
2. The impulse reaches the **AVN (atrioventricular node)** at the base of the right atrium. The AVN introduces a brief delay (~0.1 s), allowing the atria to fully empty before the ventricles contract.
3. The impulse then travels rapidly down the **Bundle of His** (in the interventricular septum) and divides into left and right bundle branches.
4. **Purkinje fibres** carry the impulse upward through the ventricular walls from the apex, ensuring the ventricles contract from the bottom upward — this squeezes blood upward and out through the arteries at the top.

The heart rate is modulated by the autonomic nervous system: the **sympathetic nerve** (releases noradrenaline) increases heart rate, while the **vagus nerve** (parasympathetic, releases acetylcholine) decreases it. The cardiovascular centre in the medulla oblongata integrates signals from chemoreceptors (detecting blood $CO_2$/pH) and baroreceptors (detecting blood pressure).

## Blood Vessels

| Vessel | Structure | Function |
|---|---|---|

| | | |
|---|---|---|
| **Arteries** | Thick muscular wall with thick elastic layer, narrow lumen relative to wall thickness, no valves, endothelium lining | Carry blood **away** from heart at high, pulsatile pressure. Thick elastic tissue stretches during systole and recoils during diastole (elastic recoil), smoothing blood flow and maintaining pressure between heartbeats. |
| **Arterioles** | Proportionally more smooth muscle than arteries; can constrict or dilate | Control blood flow to specific tissues. **Vasoconstriction** reduces blood flow (e.g. to skin in cold); **vasodilation** increases it (e.g. to muscles during exercise). Main site of peripheral resistance. |
| **Capillaries** | Wall is one cell thick (endothelium only, no muscle or elastic tissue); very narrow lumen (~8 μm, just wide enough for one red blood cell); thin basement membrane; pores between endothelial cells | Site of substance exchange between blood and tissues. One-cell-thick wall = shortest possible diffusion distance. Slow blood flow and vast total cross-sectional area maximise time for exchange. Pores allow plasma filtration (forming tissue fluid). |
| **Veins** | Thin walls with relatively little smooth muscle and elastic tissue; wide lumen; **valves** present; endothelium lining | Return blood to the heart at **low pressure**. Wide lumen offers low resistance to blood flow. Valves prevent backflow. Blood is pushed through veins by: skeletal muscle contraction (skeletal muscle pump), the respiratory pump (pressure changes in thorax during breathing), and smooth muscle in vein walls. |

**Exam Tip:** When comparing arteries and veins, always relate structure to function. Do not just list differences — explain *why* each structural feature is important. For example: "Arteries have a thick elastic layer because they must withstand and smooth out the high, pulsatile pressure of blood leaving the heart. Veins have valves because blood pressure is low and there is a risk of backflow due to gravity."

# 3.5 Water Transport in Plants

## Water Uptake and Movement Through Roots

Water enters **root hair cells** by osmosis — the soil solution has a higher water potential (less negative) than the root hair cell cytoplasm (which has a lower water potential due to dissolved mineral ions and organic solutes). Root hairs greatly increase the surface area for water absorption.

Water then moves across the cortex to the xylem via three pathways:

- **Apoplast pathway (~90% of water):** Water moves through the cell walls and intercellular spaces by mass flow, without crossing any cell membranes. This is the fastest pathway. However, it is blocked at the endodermis by the **Casparian strip**.
- **Symplast pathway:** Water passes through the cytoplasm of cells, moving from cell to cell via **plasmodesmata** (cytoplasmic connections through the cell walls). This is slower because water must cross cell membranes.
- **Vacuolar pathway:** Water passes through the vacuoles of successive cells — effectively a variation of the symplast pathway.

**Casparian strip:** A band of waxy, waterproof **suberin** in the cell walls of endodermal cells (the innermost layer of the cortex). It blocks the apoplast pathway, forcing all water and dissolved ions to pass through the cytoplasm of endodermal cells (into the symplast). This is functionally important because it allows the endodermis to act as a selective barrier — mineral ions can only enter the xylem by active transport through the endodermal cell membranes, giving the plant control over which minerals are loaded into the transpiration stream.

Mineral ions (e.g. nitrate, magnesium, potassium) are absorbed from the soil by **active transport** using carrier proteins in root hair cell membranes, because their concentration in the soil solution is usually lower than in the root cells. This requires ATP from mitochondria in root hair cells (which is why waterlogged, anaerobic soil reduces mineral uptake).

## Transpiration and the Cohesion-Tension Theory

**Transpiration:** The loss of water vapour from the aerial parts of a plant, primarily through stomata in leaves. It is a consequence of gas exchange — when stomata open for $CO_2$ uptake, water vapour inevitably escapes because the atmosphere usually has a lower water potential than the air spaces inside the leaf.

The **cohesion-tension theory** explains how water moves from roots to leaves in the xylem:

1. **Transpiration pull:** Water evaporates from the wet surfaces of mesophyll cells into the air spaces within the leaf, then diffuses out through open stomata (down a water potential gradient). This water loss lowers the water potential in the mesophyll cells.
2. **Osmosis from xylem:** The reduced water potential in mesophyll cells draws water from neighbouring cells and ultimately from the xylem vessels in the leaf veins, by osmosis.
3. **Tension (negative pressure):** Water removal from the top of the xylem creates a tension (negative pressure) that is transmitted downward through the continuous water column in the xylem.
4. **Cohesion:** Water molecules are held together by hydrogen bonds (cohesion), preventing the water column from breaking under tension. **Adhesion** between water molecules and the hydrophilic lignified xylem walls also helps support the column and counteracts gravity.
5. **Water uptake from roots:** The tension is transmitted all the way down to the roots, pulling water from the soil into the root hair cells by osmosis and through the root cortex into the xylem.

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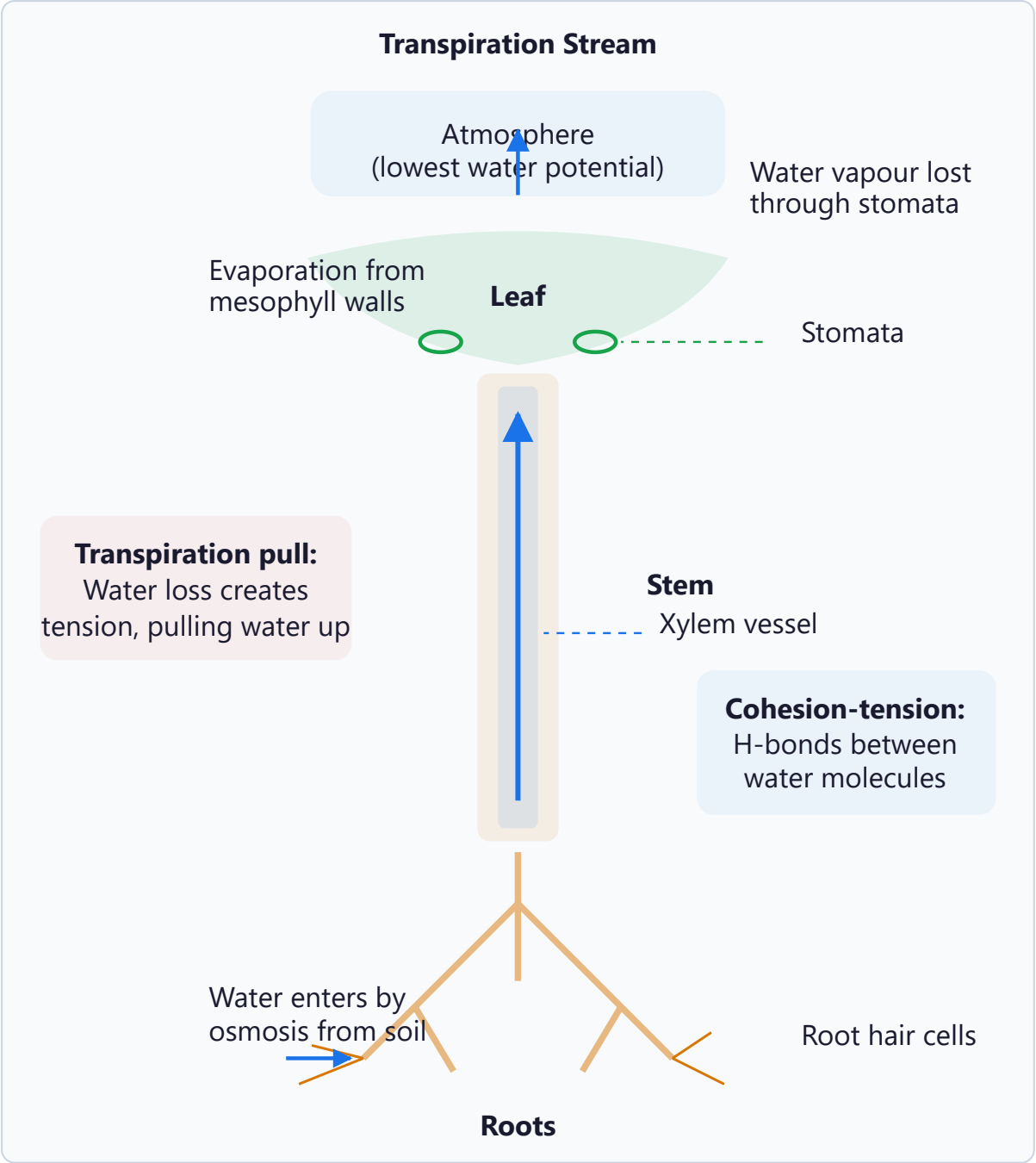


## Factors Affecting Transpiration Rate

| Factor | Effect on Rate | Explanation |
|---|---|---|
| Light intensity | Increases rate | Stomata open in light (for photosynthesis), allowing more water vapour to escape. In darkness, stomata close and transpiration is minimal. |

| | | |
|---|---|---|
| Temperature | Increases rate | Higher temperature increases kinetic energy of water molecules (faster evaporation from mesophyll surfaces), increases the water vapour capacity of air (lowering external humidity relative to the leaf), and widens the diffusion gradient. |
| Wind / air movement | Increases rate | Wind removes the layer of humid air (boundary layer) from the leaf surface, maintaining a steep concentration gradient for water vapour diffusion out of the leaf. In still air, humidity builds up around the leaf and slows diffusion. |
| Humidity | Decreases rate | High external humidity reduces the water potential gradient between the moist air spaces inside the leaf and the atmosphere outside, slowing the rate of water vapour diffusion through stomata. |
| Soil water availability | Decreases rate if low | If the soil dries out, less water is available for absorption by roots. Guard cells lose turgor and stomata close, reducing transpiration (and photosynthesis). |

**Exam Tip:** A **potometer** measures the rate of *water uptake* by a plant shoot, which is closely related to (but not exactly the same as) transpiration rate — a small proportion of water taken up is used for photosynthesis, cell expansion, and other metabolic processes. Always state this distinction in exam answers about experimental methods. Also: when setting up a potometer, ensure no air bubbles are in the system and that all joints are sealed — any leak will give inaccurate results.

## Translocation (Mass Flow in Phloem)

**Translocation:** The transport of dissolved organic solutes (assimilates) — mainly sucrose and amino acids — through the phloem sieve tubes from **sources** (where sugars are produced or mobilised, e.g. photosynthesising leaves, germinating seeds, storage organs releasing reserves) to **sinks** (where sugars are used or stored, e.g. growing roots, fruits, developing seeds, storage organs accumulating reserves).

Phloem structure:

- **Sieve tube elements:** Living cells arranged end-to-end, connected by **sieve plates** (perforated end walls that allow mass flow of phloem sap). They lack a nucleus, ribosomes, and most organelles to maximise space for flow.
- **Companion cells:** Metabolically active cells adjacent to sieve tube elements, connected by many plasmodesmata. They have dense cytoplasm, many mitochondria (for ATP), and a nucleus. They carry out the metabolic functions for the sieve tube elements and are responsible for actively loading sucrose into the phloem.

The **mass flow (pressure flow) hypothesis** explains translocation:

1. **Loading at the source:** Sucrose is actively loaded into sieve tube elements at the source. Companion cells use ATP to pump $H^+$ ions out of the cell (via a proton pump). $H^+$ ions then flow back in through sucrose-$H^+$ co-transporters, carrying sucrose with them against its concentration gradient. Sucrose then passes from companion cells into sieve tube elements through plasmodesmata. This active loading lowers the water potential (Ψ) in the sieve tube at the source.
2. Water enters the sieve tube by **osmosis** from nearby xylem vessels (down the water potential gradient), increasing the **hydrostatic pressure** at the source end.
3. **Unloading at the sink:** Sucrose is actively or passively unloaded from sieve tube elements into sink cells (e.g. for respiration, conversion to starch for storage, or incorporation into cellulose for cell walls). Removing sucrose raises the water potential in the sieve tube at the sink.
4. Water leaves the sieve tube at the sink by osmosis (back into the xylem), **lowering** hydrostatic pressure at the sink end.
5. The **pressure difference** between source (high pressure) and sink (low pressure) drives the bulk flow (mass flow) of phloem sap from source to sink through the sieve plates.

Evidence supporting the mass flow hypothesis includes: phloem sap flows from high to low pressure (shown by aphid stylet experiments and pressure measurements), radioactive tracer studies showing movement from source to sink, and the requirement for ATP (metabolic inhibitors block loading). Evidence against includes: sucrose and amino acids travel at different rates (unexpected for mass flow of a single solution) and sieve plates present resistance to flow.

**Exam Tip:** Do not confuse xylem transport with phloem transport. Xylem carries water and mineral ions *upward* only (unidirectional), driven by transpiration pull (passive, no ATP). Phloem carries organic solutes in *any direction* (bidirectional, source to sink — which can be upward or downward

depending on where the source and sink are), driven by a pressure gradient created by active loading (requires ATP). A common exam question asks you to compare the two systems in a table.